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POSITIVE ELECTRODE ACTIVE MATERIAL AND PREPARATION METHOD THEREFOR, SECONDARY BATTERY, BATTERY MODULE, BATTERY PACK AND ELECTRIC DEVICE

Abstract

A positive electrode active material, a secondary battery, a battery module, a battery pack, and an electric device. The positive electrode active material is used as a positive electrode active material for a secondary battery, and comprises a carbon material compounded iron-based polyanionic compound and an aluminum-containing oxide, and the iron-based polyanionic compound has the following general formula: $\text{Na}_{.4}\text{Fe}_{.3-x}\text{M}_{.x}\text{Al}_{.y}(\text{PO}_{.4})_{.2}\text{P}_{.2}\text{O}_{.7}/\text{C}$, wherein M comprises a transition metal element, $0 \leq x \leq 0.5$, and y is greater than 0 and less than 0.2. The positive electrode active material has relatively low residual alkali amount, and the battery has excellent cycle performance and rate capability.

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Background/Summary

CROSS-REFERENCE OF RELATED APPLICATIONS [0001] The present application is a continuation of PCT Application No. PCT/CN2023/124961, filed on Oct. 17, 2023, which claims priority to Chinese Patent Application No. 202310075495.3, entitled “POSITIVE ELECTRODE ACTIVE MATERIAL AND PREPARATION METHOD THEREFOR, SECONDARY BATTERY, BATTERY MODULE, BATTERY PACK AND ELECTRIC DEVICE”, filed on Jan. 16, 2023, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present application relates to the technical field of secondary batteries, and in particular, to a positive electrode active material and a preparation method therefor, a secondary battery, a battery module, a battery pack, and an electric device.

BACKGROUND

[0003] Sodium batteries have great application potential in large-scale energy storage due to the abundant reserves of sodium, low prices, and wide operating temperature range.

[0004] As a popular positive electrode material for novel sodium batteries, iron-based polyanionic compounds have significant advantages such as abundant resources, high operating voltage, good cycle stability, and environmental friendliness. However, sodium batteries with iron-based polyanionic compounds as positive electrode materials still face issues such as low electronic conductivity, low discharge capacity, and poor cycle performance. These issues make them unable to meet the application requirements of the new generation of electrochemical systems.

SUMMARY

[0005] The present application is made in view of the issues described above, and its purpose is to provide a positive electrode active material, in which an aluminum element is introduced to achieve the dual effects of lattice doping and composite modification, thereby reducing the residual alkali content of the positive electrode active material and improving the cycle performance and rate capability of the battery.

[0006] In a first aspect of the present application, provided is a positive electrode active material for use in a secondary battery, which includes a carbon material-composited iron-based polyanionic compound and an aluminum-containing oxide, where the iron-based polyanionic compound has the following general formula: $\text{Na}_{0.4}\text{Fe}_{0.3-x}\text{M}_x\text{Al}_y(\text{PO}_4)_2\text{P}_2\text{O}_7/\text{C}$, where M includes a transition metal element, $0 \leq x \leq 0.5$, and $0 < y < 0.2$.

[0007] By introducing an aluminum element into the positive electrode active material, in one aspect, a portion of the aluminum element can be doped into the lattice of the iron-based polyanionic compound to replace the iron element or the transition metal element in the lattice of the iron-based polyanionic compound, thereby enhancing the ionic conductivity and electronic conductivity of the positive electrode active material, and improving ion-conducting and electron-

conducting properties of the positive electrode active material. In another aspect, a portion of the aluminum element is distributed on the surface of the iron-based polyanionic compound in the form of an aluminum-containing oxide, which reduces the direct contact between the surface of the positive electrode active material and the moisture in the environment, lowers the residual alkali content of the positive electrode active material, and improves the stability and processability of the positive electrode active material. Moreover, the positive electrode active material also has a high specific capacity, which is conducive to improving the energy density of the battery. By introducing a carbon material into the positive electrode active material, the electrical conductivity of the positive electrode active material and battery performance can be effectively improved.

[0008] In any embodiment, the aluminum-containing oxide is distributed on at least a portion of the surface of a primary particle of the iron-based polyanionic compound.

[0009] The aluminum-containing oxide is distributed on either a portion of the surface or the entire surface of a primary particle of the iron-based polyanionic compound, which can effectively reduce the direct contact between the surface of the positive electrode active material and the moisture in the environment, lower the residual alkali content of the positive electrode active material, and improve the stability and processability of the positive electrode active material.

[0010] In any embodiment, the mass content of the aluminum element in the positive electrode active material is 0.03% to 0.9% based on the total mass of the positive electrode active material.

[0011] The mass content of the aluminum element in the positive electrode active material is controlled to be 0.03% to 0.9% based on the total mass of the positive electrode active material, which is conducive to enhancing the electrical conductivity of the positive electrode active material, reducing the residual alkali content on the surface of the material, and improving the stability and processability of the positive electrode active material, thereby increasing the specific capacity of the positive electrode active material and the capacity retention rate of the battery at 3C rate and improving the capacity and rate capability of the battery.

[0012] In any embodiment, M includes one or more of Ni, Co, Mn, Cu, V, Ti, Mo, Nb, W, Cr, Zn, Zr, and Ca, optionally one or more of Ni, Co, Mn, Cu, V, and Ca.

[0013] The doping of transition metal elements into an iron-based polyanionic compound is conducive to increasing the voltage plateau of the positive electrode active material, improving the ionic conductivity and electronic conductivity of the material, and reducing battery polarization, thereby improving the cycle performance and rate capability of the battery.

[0014] In any embodiment, M includes at least two of Ni, Co, Mn, Cu, V, and Ca.

[0015] M is controlled to include at least two of Ni, Co, Mn, Cu, V, and Ca, which is conducive to further improving the rate capability of the battery.

[0016] In any embodiment, the surface of the iron-based polyanionic compound is coated with the carbon material in the form of a carbon film, or the carbon material is distributed between primary particles of the iron-based polyanionic compound in the form of particles.

[0017] The surface of the iron-based polyanionic compound is coated with the carbon material in the form of a carbon film, or the carbon material is distributed between primary particles of the iron-based polyanionic compound in the form of particles, either of which can effectively improve the electrical conductivity of the positive electrode active material and the battery performance.

[0018] In any embodiment, the carbon material includes one or more of amorphous carbon, conductive carbon black, a carbon nanotube, and graphene.

[0019] The carbon material described above is readily dispersible and easy to process, and forms a carbon composite with an iron-based polyanionic compound, which improves the electrical conductivity of the positive electrode active material and the battery performance. Since the crystallinity of conductive carbon black, a carbon nanotube, or graphene is higher than that of amorphous carbon, the electrical conductivity of the carbon material including conductive carbon black, a carbon nanotube, or graphene is better than that of the carbon material including amorphous carbon.

[0020] In any embodiment, the mass content of the carbon material is 0.5% to 6%, optionally 1% to 3.6%, based on the total mass of the positive electrode active material.

[0021] The mass content of the carbon material is controlled to be 0.5% to 6% based on the total mass of the positive electrode active material, which is conducive to achieving a balance between the specific capacity and the electrical conductivity of the positive electrode active material, while also attaining a higher specific capacity and better cycle performance and rate capability. It should be avoided that the mass content of the carbon material is excessively low, which leads to poor electrical conductivity of the positive electrode active material and thus adversely affects the capacity volatilization of the positive electrode active material; it should be also avoided that the mass content of the carbon material is excessively high, which leads to a decrease in the specific capacity of the positive electrode active material. The mass content of the carbon material is controlled to be 1% to 3.6% based on the total mass of the positive electrode active material, which is conducive to further increasing the specific capacity of the positive electrode active material, thereby further improving the capacity of the battery.

[0022] In any embodiment, the specific capacity of the positive electrode active material is not less than 98 mAh/g.

[0023] A positive electrode active material with a specific capacity of not less than 98 mAh/g is conducive to improving the capacity of the battery, thereby broadening the application of the battery.

[0024] In any embodiment, the residual alkali content of NaHCO_3 on the surface of the positive electrode active material is less than 1.2% based on the total mass of the positive electrode active material.

[0025] The residual alkali content of NaHCO_3 on the surface of the positive electrode active material is less than 1.2% based on the total mass of the positive electrode active material, which can effectively improve the stability and processability of the positive electrode active material.

[0026] In a second aspect of the present application, provided is a preparation method for a positive electrode active material for use in a secondary battery. The method includes the following steps:

[0027] dissolving raw materials including an iron source, a sodium source, a phosphorus source, an aluminum source, and a carbon source in water to obtain a mixed slurry, where the raw materials optionally include an M source, and the M source is a salt containing a transition metal; and [0028] drying and then calcining the mixed slurry to prepare the positive electrode active material including an iron-based polyanionic compound and an aluminum-containing oxide, where the iron-based polyanionic compound has the following general formula:

$\text{Na}_{0.4}\text{Fe}_{0.3-x}\text{M}_x\text{Al}_y(\text{PO}_4)_2\text{P}_2\text{O}_7/\text{C}$,
where M includes a transition metal element, $0 \leq x \leq 0.5$, and $0 < y < 0.2$.

[0029] The preparation method for the positive electrode active material described above is simple and the production cost is low. The prepared positive electrode active material has a relatively low residual alkali content and a high specific capacity, and the battery has excellent cycle performance and rate capability.

[0030] In some embodiments, $0 < x < 0.5$.

[0031] The introduction of transition metals into the positive electrode active material is conducive to improving the ionic conductivity and electronic conductivity of the material, thereby improving the cycle performance and rate capability of the battery.

[0032] In any embodiment, drying and then calcining the mixed slurry includes the following steps:

[0033] drying the mixed slurry to obtain a precursor powder; and [0034] subjecting the precursor powder to a stepwise calcination process, with the temperature of the first calcination step being 300°C . to 400°C . and the time being 3 h to 6 h, and the temperature of the second calcination step being 500°C . to 600°C . and the time being 8 h to 15 h, to prepare the positive electrode active material.

[0035] Stepwise calcination is conducive to the complete reaction of the precursor powder, which enhances the crystallinity and phase purity of the prepared positive electrode active material, reduces the residual alkali content on the surface of the positive electrode active material, improves the stability and processability of the positive electrode active material, and increases the specific capacity of the positive electrode active material, as well as the cycle capacity retention rate and capacity retention rate at 3C rate of the battery, thereby improving the capacity, cycle performance, and rate capability of the battery.

[0036] In any embodiment, the temperature of the second calcination step is 525° C. to 575° C., optionally 550° C. to 575° C.

[0037] The temperature of the second calcination step is controlled to be 525° C. to 575° C., which is conducive to further reducing the residual alkali content on the surface of the active material and improving the electrical conductivity of the material, thereby improving the cycle performance and rate capability of the battery. The temperature of the second calcination step is further controlled to be 550° C. to 575° C., which is conducive to increasing the capacity retention rate after 200 cycles and the capacity retention rate at 3C rate of the battery to a greater extent, thereby improving the cycle performance and rate capability of the battery to a greater extent.

[0038] In any embodiment, the time of the second calcination step is 10 h to 14 h.

[0039] The time of the second calcination step is controlled to be 10 h to 14 h, which is conducive to obtaining a positive electrode active material with high crystallinity and high purity. At the same time, by controlling the holding time, the particle size of the primary crystal grains and the residual alkali content on the surface of the material can be controlled, thereby improving the specific capacity of the material and the cycle rate capability of the battery.

[0040] In any embodiment, the M source includes one or more of nickel nitrate, nickel acetate, cobalt nitrate, manganese nitrate, cobalt acetate, manganese acetate, nickel oxalate, cobalt oxalate, manganese oxalate, nickel oxide, cobalt oxide, and manganese oxide.

[0041] In any embodiment, the aluminum source includes one or more of aluminum nitrate, aluminum acetylacetonate, aluminum acetate, aluminum hydroxide, and aluminum oxide.

[0042] In any embodiment, the carbon source includes one or more of sucrose, tannic acid, polyethylene glycol, polyacrylonitrile, cellulose, polyvinylpyrrolidone, sucrose, oxalic acid, glucose, ascorbic acid, polyethylene, citric acid, conductive carbon black, a carbon nanotube, and graphene.

[0043] The carbon source described above, when calcined at high temperatures, readily generates a carbon material that either coats the surface of the iron-based polyanionic compound or is distributed between primary particles of the iron-based polyanionic compound, thereby improving the electrical conductivity of the positive electrode active material. Due to the influence of the calcination temperature, the amorphous carbon formed by the carbon source including one or more of sucrose, tannic acid, polyethylene glycol, polyacrylonitrile, cellulose, polyvinylpyrrolidone, sucrose, oxalic acid, glucose, ascorbic acid, polyethylene, citric acid, conductive carbon black, a carbon nanotube, and graphene has relatively low crystallinity, while the carbon material generated by the carbon source including one or more of conductive carbon black, a carbon nanotube, and graphene has relatively high crystallinity. Therefore, the carbon material generated by the carbon source including one or more of conductive carbon black, a carbon nanotube, and graphene has more excellent conductive property.

[0044] In a third aspect of the present application, provided is a secondary battery, which includes a positive electrode plate. The positive electrode plate includes the positive electrode active material according to any embodiment or the positive electrode active material prepared by the preparation method according to any embodiment.

[0045] In any embodiment, the secondary battery is a negative electrode-free sodium battery.

[0046] In any embodiment, the secondary battery further includes a negative electrode plate, the negative electrode plate includes a negative electrode current collector and a bottom coating

arranged on at least one surface of the negative electrode current collector, and the bottom coating includes one or more of a carbon nanotube, graphite, graphene, a silver-carbon composite nanoparticle (Ag/C), and a tin-carbon composite nanoparticle (Sn/C).

[0047] The bottom coating described above not only has excellent electrical conductivity, but is also conducive to the uniform deposition of metal ions on the surface of the current collector, thereby improving the cycle performance and safety of the battery.

[0048] In any embodiment, the surface density of the bottom coating is 5 g/m.² to 50 g/m.².

[0049] The bottom coating with a surface density of 5 g/m.² to 50 g/m.² is conducive to the uniform distribution of nucleation sites, promotes the uniform deposition of metals, and does not affect the transmission of electrons.

[0050] In any embodiment, the thickness of the bottom coating is 2 μm to 100 μm.

[0051] The thickness of the bottom coating is controlled to be 2 μm to 100 μm, and thus sufficient nucleation sites can be provided, which is conducive to the uniform deposition of metal ions, thereby inhibiting dendrite formation.

[0052] In a fourth aspect of the present application, provided is a battery module, which includes the secondary battery according to the third aspect of the present application.

[0053] In a fifth aspect of the present application, provided is a battery pack, which includes the secondary battery according to the third aspect of the present application or the battery module according to the fourth aspect of the present application.

[0054] In a sixth aspect of the present application, provided is an electric device, which includes at least one of the secondary battery according to the third aspect of the present application, the battery module according to the fourth aspect of the present application, and the battery pack according to the fifth aspect of the present application.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0055] FIG. 1 is a schematic diagram of a secondary battery according to one embodiment of the present application.

[0056] FIG. 2 is an exploded view of the secondary battery according to one embodiment of the present application as shown in FIG. 1.

[0057] FIG. 3 is a schematic diagram of a battery module according to one embodiment of the present application.

[0058] FIG. 4 is a schematic diagram of a battery pack according to one embodiment of the present application.

[0059] FIG. 5 is an exploded view of the battery pack according to one embodiment of the present application as shown in FIG. 4.

[0060] FIG. 6 is a schematic diagram of an electric device using a secondary battery as a power source according to one embodiment of the present application.

DESCRIPTION OF THE REFERENCE NUMERALS

[0061] 1: battery pack; 2: upper case body; 3: lower case body; 4: battery module; 5: secondary battery; 51: housing; 52: electrode assembly; 53: cover plate.

DETAILED DESCRIPTION

[0062] Hereinafter, embodiments specifically disclosing the positive electrode active material and the preparation method therefor, the secondary battery, the battery module, the battery pack, and the electric device of the present application are described in detail with appropriate reference to the drawings. However, unnecessarily detailed descriptions may be omitted. For example, detailed descriptions of well-known matters and repetitive descriptions of actually identical structures may

be omitted. This is to avoid unnecessary lengthiness of the following descriptions and to facilitate understanding by those skilled in the art. Additionally, the drawings and the following descriptions are provided to enable those skilled in the art to fully understand the present application and are not intended to limit the subject matter recited in the claims.

[0063] The “ranges” disclosed in the present application are defined with lower and upper limits. A given range is defined by selecting a lower limit and an upper limit that delineate the boundaries of a particular range. Ranges defined in this manner may include or exclude the end values and can be combined arbitrarily, which means that any lower limit may be combined with any upper limit to form a range. For example, if ranges of 60-120 and 80-110 are listed for a particular parameter, it is understood that ranges of 60-110 and 80-120 are also anticipated. Additionally, if the minimum range values listed are 1 and 2, and the maximum range values listed are 3, 4, and 5, then the following ranges can all be anticipated: 1-3, 1-4, 1-5, 2-3, 2-4, and 2-5. In the present application, unless otherwise specified, the numerical range “a-b” indicates an abbreviated representation of any combination of real numbers between a and b, where both a and b are real numbers. For example, the numerical range “0-5” indicates that all real numbers between “0-5” are listed herein, and “0-5” is merely an abbreviated representation of a combination of these numerical values. Additionally, when stating that a parameter is an integer ≥ 2 , it is equivalent to disclosing that the parameter is, for example, an integer of 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or the like.

[0064] Unless otherwise specified, all embodiments and optional embodiments of the present application can be combined with one another to form new technical solutions.

[0065] Unless otherwise specified, all technical features and optional technical features of the present application can be combined with one another to form new technical solutions.

[0066] Unless otherwise specified, all steps of the present application can be performed sequentially or randomly, preferably sequentially. For example, if the method includes steps (a) and (b), it indicates that the method may include steps (a) and (b) performed sequentially or steps (b) and (a) performed sequentially. For example, if the mentioned method may further include step (c), it indicates that step (c) may be added to the method in any order; for example, the method may include steps (a), (b), and (c), or steps (a), (c), and (b), or steps (c), (a), and (b), or the like.

[0067] Unless otherwise specified, the “include” and “comprise” mentioned in the present application are open-ended or closed-ended. For example, the “include” and “comprise” may mean that other unlisted components may also be included or comprised or that only the listed components are included or comprised.

[0068] Unless otherwise specified, the term “or” in the present application is inclusive. For example, the phrase “A or B” means “A, B, or both A and B”. More specifically, any one of the following conditions satisfies the condition “A or B”: A is true (or present) and B is false (or absent); A is false (or absent) and B is true (or present); or both A and B are true (or present).

[0069] In the prior art, polyanionic compounds have become one of the popular positive electrode materials for sodium batteries due to their abundant resources, environmental friendliness, convenience for large-scale production, open sodium ion diffusion channels, and good thermal stability and cycle stability. However, in the actual production process, researchers have found that the polyanionic compound material exhibits a high residual alkali content on the surface, which tends to increase when the material is exposed to air, causing deterioration in specific capacity and thus affecting the performance of the battery. Therefore, it is necessary to develop a positive electrode active material with a low residual alkali content and excellent electrochemical performance, which can be used for high-rate batteries to meet the application requirements of the new generation of electrochemical systems.

Positive Electrode Active Material

[0070] Based on this, the present application provides a positive electrode active material for use in a secondary battery. The positive electrode active material includes a carbon material-composited iron-based polyanionic compound and an aluminum-containing oxide, where the iron-based

polyanionic compound has the following general formula:

$\text{Na}_{0.4}\text{Fe}_{0.3-x}\text{M}_x\text{Al}_y(\text{PO}_4)_2\text{P}_2\text{O}_7/\text{C}$, where M includes a transition metal element, $0 \leq x \leq 0.5$, and $0 < y < 0.2$.

[0071] The term “transition metal element” used herein refers to the elements of Groups IIIB to VIIB and VIII, and Groups IB to IIB in the Periodic Table of the Elements.

[0072] In some embodiments, x is optionally 0, 0.05, 0.1, 0.15, 0.2, 0.25, 0.3, 0.35, 0.4, 0.45, or 0.5.

[0073] In some embodiments, y is optionally 0.02, 0.04, 0.06, 0.08, 0.10, 0.12, 0.14, 0.16, 0.18, or 0.19.

[0074] By introducing an aluminum element into the positive electrode active material, in one aspect, a portion of the aluminum element can be doped into the lattice of the iron-based polyanionic compound to replace the iron element or the transition metal element in the lattice of the iron-based polyanionic compound, thereby enhancing the electronic conductivity of the positive electrode active material, and improving the electron-conducting property of the positive electrode active material. In another aspect, a portion of the aluminum element is distributed on the surface of the iron-based polyanionic compound in the form of an aluminum-containing oxide, which reduces the direct contact between the surface of the positive electrode active material and the moisture in the environment, lowers the residual alkali content of the positive electrode active material, and improves the stability and processability of the positive electrode active material. Moreover, the positive electrode active material also has a high specific capacity, which is conducive to improving the energy density of the battery. By introducing a carbon material into the positive electrode active material, the electrical conductivity of the positive electrode active material and battery performance can be effectively improved.

[0075] The term “specific capacity” used herein refers to the amount of electricity released per gram of the positive electrode active material, which can reflect the capacity of the battery and can be tested by any known method.

[0076] In some embodiments, the aluminum-containing oxide is distributed on at least a portion of the surface of a primary particle of the iron-based polyanionic compound.

[0077] In some embodiments, the aluminum-containing oxide includes aluminum oxide.

[0078] The term “primary particle” used herein refers to particles of the iron-based polyanionic compound before agglomeration, and the particle size of the primary particles ranges from 30 nm to 120 nm.

[0079] In some embodiments, the aluminum-containing oxide is distributed on a portion of the surface of a primary particle of the iron-based polyanionic compound.

[0080] In some embodiments, the aluminum-containing oxide is distributed on the entire surface of a primary particle of the iron-based polyanionic compound.

[0081] The aluminum-containing oxide is distributed on either a portion of the surface or the entire surface of a primary particle of the iron-based polyanionic compound, which can effectively reduce the direct contact between the surface of the positive electrode active material and the moisture in the environment, lower the residual alkali content of the positive electrode active material, and improve the stability and processability of the positive electrode active material.

[0082] In some embodiments, the mass content of the aluminum element in the positive electrode active material is 0.03% to 0.9% based on the total mass of the positive electrode active material. In some embodiments, the mass content of the aluminum element in the positive electrode active material is optionally 0.03%, 0.05%, 0.1%, 0.15%, 0.2%, 0.25%, 0.3%, 0.35%, 0.4%, 0.45%, 0.5%, 0.55%, 0.6%, 0.65%, 0.7%, 0.75%, 0.8%, 0.85%, or 0.9% based on the total mass of the positive electrode active material.

[0083] In some embodiments, the aluminum element in the positive electrode active material includes the aluminum element in the iron-based polyanionic compound and the aluminum element in the aluminum-containing oxide.

[0084] The mass content of the aluminum element in the positive electrode active material can be tested by any means known in the art. As an example, reference can be made to the standard YS/T 1006.2-2014, GB/T 23367.2-2009, or YS/T 1028.5-2015. Specifically, an inductively coupled plasma optical emission spectrometer (Thermo, ICAP7400; PerkinElmer, Avio200) can be used to measure the emission intensity of the aluminum element in the test sample. Standard samples with different contents of aluminum element are prepared, and their emission intensities are measured to establish a standard curve. The mass content of the aluminum element in the test sample is quantified by matching its emission intensity to the standard curve.

[0085] The mass content of the aluminum element in the positive electrode active material is controlled to be 0.03% to 0.9% based on the total mass of the positive electrode active material, which is conducive to reducing the residual alkali content of the positive electrode active material and improving the stability and processability of the positive electrode active material, thereby increasing the specific capacity of the positive electrode active material and the capacity retention rate of the battery at 3C rate and improving the capacity and rate capability of the battery.

[0086] In some embodiments, M includes one or more of Ni, Co, Mn, Cu, V, Ti, Mo, Nb, W, Cr, Zn, Zr, and Ca, optionally one or more of Ni, Co, Mn, Cu, V, and Ca.

[0087] In some embodiments, M includes Ni. In some embodiments, M includes Co. In some embodiments, M includes Mn. In some embodiments, M includes Ca. In some embodiments, M includes Cu.

[0088] The doping of transition metal elements into an iron-based polyanionic compound is conducive to improving the electronic conductivity and ionic conductivity of the positive electrode active material and reducing battery polarization, thereby improving the cycle performance and rate capability of the battery.

[0089] In some embodiments, M includes at least two of Ni, Co, Mn, Cu, V, and Ca.

[0090] In some embodiments, M includes Ni and Mn. In some embodiments, M includes Mn and Co. In some embodiments, M includes Ni and Co. In some embodiments, M includes Ni and Ca. In some embodiments, M includes Mn, Ni, and Co.

[0091] M is controlled to include at least two of Ni, Co, Mn, Cu, V, and Ca, which is conducive to further improving the rate capability of the battery.

[0092] In some embodiments, the surface of the iron-based polyanionic compound is coated with the carbon material in the form of a carbon film, or the carbon material is distributed between primary particles of the iron-based polyanionic compound in the form of particles.

[0093] In some embodiments, the surface of the iron-based polyanionic compound is coated with the carbon material in the form of a carbon film.

[0094] In some embodiments, the carbon material is distributed between primary particles of the iron-based polyanionic compound in the form of particles.

[0095] The surface of the iron-based polyanionic compound is coated with the carbon material in the form of a carbon film, or the carbon material is distributed between primary particles of the iron-based polyanionic compound in the form of particles, either of which can effectively improve the electrical conductivity of the positive electrode active material and the battery performance.

[0096] In some embodiments, the carbon material includes one or more of amorphous carbon, conductive carbon black, a carbon nanotube, and graphene.

[0097] In some embodiments, the carbon material includes amorphous carbon. In some embodiments, the carbon material includes conductive carbon black. In some embodiments, the carbon material includes a carbon nanotube. In some embodiments, the carbon material includes graphene.

[0098] The carbon material described above readily composites with the iron-based polyanionic compound, and thus the conductive property of the positive electrode active material is improved. Since the crystallinity of conductive carbon black, a carbon nanotube, or graphene is higher than that of amorphous carbon, the electrical conductivity of the carbon material including conductive

carbon black, a carbon nanotube, or graphene is better than that of the carbon material including amorphous carbon.

[0099] In some embodiments, the mass content of the carbon material is 0.5% to 6%, optionally 1% to 3.6%, based on the total mass of the positive electrode active material. In some embodiments, the mass content of the carbon material is optionally 0.5%, 1%, 1.5%, 2%, 2.5%, 3%, 3.5%, 3.6%, 4%, 4.5%, 5%, 5.5%, or 6% based on the total mass of the positive electrode active material.

[0100] The mass content of the carbon material is controlled to be 0.5% to 6% based on the total mass of the positive electrode active material, which is conducive to improving the electrical conductivity of the positive electrode active material, while ensuring high specific capacity performance, thereby improving the capacity, cycle performance, and rate capability of the battery. It should be avoided that the mass content of the carbon material is excessively low, which leads to poor electrical conductivity of the positive electrode active material and thus adversely affects the capacity performance of the positive electrode active material; it should be also avoided that the mass content of the carbon material is excessively high, which leads to a decrease in the specific capacity of the positive electrode active material. The mass content of the carbon material is controlled to be 1% to 3.6% based on the total mass of the positive electrode active material, which is conducive to further increasing the specific capacity of the positive electrode active material, thereby further improving the capacity of the battery.

[0101] In some embodiments, the specific capacity of the positive electrode active material is not less than 98 mAh/g.

[0102] In some embodiments, the specific capacity of the positive electrode active material is optionally 98 mAh/g, 100 mAh/g, 105 mAh/g, 110 mAh/g, 115 mAh/g, 120 mAh/g, 125 mAh/g, or 129 mAh/g.

[0103] The specific capacity of the positive electrode active material can be tested by any means known in the art. As an example, at 25° C. and normal pressure, a button battery is charged at a constant current rate of 0.1 C to a voltage of 4 V, and then charged at a constant voltage of 4 V until the current drops to 0.05 C, and the charge-specific capacity at this time is recorded as the first desodiation capacity. Subsequently, the battery is discharged at a constant current rate of 0.1 C to a voltage of 1.5 V, and the discharge-specific capacity at this time is recorded as the first sodiation capacity. The specific capacity of the positive electrode active material is the first sodiation capacity.

[0104] A positive electrode active material with a specific capacity of not less than 98 mAh/g is conducive to improving the capacity of the battery, thereby broadening the application of the battery.

[0105] In some embodiments, the residual alkali content of NaHCO₃ on the surface of the positive electrode active material is less than 1.2% based on the total mass of the positive electrode active material.

[0106] In some embodiments, the residual alkali content of NaHCO₃ on the surface of the positive electrode active material is optionally 0.01%, 0.05%, 0.1%, 0.2%, 0.3%, 0.4%, 0.5%, 0.6%, 0.7%, 0.8%, 0.9%, 1%, 1.1%, or 1.15% based on the total mass of the positive electrode active material.

[0107] The residual alkali content of NaHCO₃ in the positive electrode active material can be tested by any means known in the art. As an example, the residual alkali content is determined through a potentiometric titration method using chemical reagents. For example, reference can be made to the standard GB/T 9725-2007. Specifically, a Metrohm 905 Titrando potentiometric titrator can be used. The positive electrode active material is dissolved in deionized water, and the aqueous positive electrode active material solution is titrated with a standard titration solution. For each addition of 0.1 mL of the standard titration solution, the potential or pH value is recorded. The titration is stopped when the potential or pH value shows slight variation. The volume of the added

standard titration solution and the measured potential or pH value are recorded, the titration end point is determined using a graphical method or the second derivative method, and the titration volume of the standard titration solution is determined. The mass of NaHCO_3 obtained through calculation is divided by the mass of the positive electrode active material, and the mass content is taken as the residual alkali content of NaHCO_3 in the positive electrode active material.

[0108] The residual alkali content of NaHCO_3 in the positive electrode active material is less than 1.2% based on the total mass of the positive electrode active material, which can effectively improve the stability and processability of the positive electrode active material.

[0109] The present application further provides a preparation method for a positive electrode active material for use in a secondary battery. The method includes the following steps: [0110] dissolving raw materials including an iron source, a sodium source, a phosphorus source, and an aluminum source in water to obtain a mixed slurry, where the raw materials optionally include an M source, and the M source is a salt containing a transition metal; and [0111] drying and then calcining the mixed slurry to prepare the positive electrode active material including an iron-based polyanionic compound and an aluminum-containing oxide, where the iron-based polyanionic compound has the following general formula:

$\text{Na}_{0.4}\text{Fe}_{0.3-x}\text{M}_x\text{Al}_y(\text{PO}_4)_2\text{P}_2\text{O}_7/\text{C}$,
where M includes a transition metal element, $0 \leq x \leq 0.5$, and $0 < y < 0.2$.

[0112] The preparation method for the positive electrode active material described above is simple and the production cost is low. The prepared positive electrode active material has a relatively low residual alkali content and a high specific capacity, and the battery has excellent cycle performance and rate capability.

[0113] In some embodiments, $0 < x \leq 0.5$.

[0114] The introduction of transition metals into the positive electrode active material is conducive to improving the ionic conductivity and electronic conductivity of the material, thereby improving the cycle performance and rate capability of the battery.

[0115] In some embodiments, drying and then calcining the mixed slurry includes the following steps: [0116] drying the mixed slurry to obtain a precursor powder; and [0117] subjecting the precursor powder to a stepwise calcination process, with the temperature of the first calcination step being 300° C. to 400° C. and the time being 3 h to 6 h, and the temperature of the second calcination step being 500° C. to 600° C. and the time being 8 h to 15 h, to prepare the positive electrode active material.

[0118] Stepwise calcination is conducive to the complete reaction of the precursor powder, which enhances the crystallinity and phase purity of the prepared positive electrode active material, reduces the residual alkali content on the surface of the positive electrode active material, improves the stability and processability of the positive electrode active material, and increases the specific capacity of the positive electrode active material, thereby improving the cycle performance and rate capability of the battery.

[0119] In some embodiments, the temperature of the second calcination step is 525° C. to 575° C., optionally 550° C. to 575° C.

[0120] The temperature of the second calcination step is controlled to be 525° C. to 575° C., which is conducive to further reducing the residual alkali content on the surface of the active material and improving the electrical conductivity of the material, thereby improving the cycle performance and rate capability of the battery. The temperature of the second calcination step is further controlled to be 550° C. to 575° C., which is conducive to improving the cycle performance and rate capability of the battery to a greater extent.

[0121] In some embodiments, the time of the second calcination step is 10 h to 14 h.

[0122] The time of the second calcination step is controlled to be 10 h to 14 h, which is conducive

to obtaining a positive electrode active material with high crystallinity and high purity. At the same time, by controlling the holding time, the particle size of the primary crystal grains and the residual alkali content on the surface of the material can be controlled, thereby improving the specific capacity of the material and the cycle rate capability of the battery.

[0123] In some embodiments, the M source includes one or more of nickel nitrate, nickel acetate, cobalt nitrate, manganese nitrate, cobalt acetate, manganese acetate, nickel oxalate, cobalt oxalate, manganese oxalate, nickel oxide, cobalt oxide, and manganese oxide.

[0124] In some embodiments, the aluminum source includes one or more of aluminum nitrate, aluminum acetylacetonate, aluminum acetate, aluminum hydroxide, and aluminum oxide.

[0125] In some embodiments, the carbon source includes one or more of sucrose, tannic acid, polyethylene glycol, polyacrylonitrile, cellulose, polyvinylpyrrolidone, sucrose, oxalic acid, glucose, ascorbic acid, polyethylene, citric acid, conductive carbon black, a carbon nanotube, and graphene.

[0126] In some embodiments, the carbon source includes one or more of sucrose, tannic acid, polyethylene glycol, polyacrylonitrile, cellulose, polyvinylpyrrolidone, sucrose, oxalic acid, glucose, ascorbic acid, polyethylene, and citric acid.

[0127] In some embodiments, the carbon source includes one or more of conductive carbon black, a carbon nanotube, and graphene.

[0128] The carbon source described above, when calcined at high temperatures, readily generates a carbon material that either coats the surface of the iron-based polyanionic compound or is distributed between primary particles of the iron-based polyanionic compound, thereby improving the electrical conductivity of the positive electrode active material. Due to the influence of the calcination temperature, the amorphous carbon formed by the organic carbon source has relatively low crystallinity, while the carbon material generated by the carbon source including one or more of conductive carbon black, a carbon nanotube, graphene, and other inorganic carbon sources has relatively high crystallinity. Therefore, the carbon material generated by the carbon source including one or more of conductive carbon black, a carbon nanotube, and graphene has more excellent conductive property.

Positive Electrode Plate

[0129] A positive electrode plate includes a positive electrode current collector and a positive electrode active material layer formed on at least a portion of the surface of the positive electrode current collector, where the positive electrode active material layer includes the positive electrode active material according to some embodiments.

[0130] The positive electrode active material layer may also include a conductive agent to improve the conductive property of the positive electrode. The conductive agent is optionally one or more of super P, acetylene black, carbon black, Ketjen black, a carbon dot, a carbon nanotube, graphite, graphene, and a carbon nanofiber.

[0131] The positive electrode active material layer may also include a binder to firmly bond the positive electrode active material and the optional conductive agent to the positive electrode current collector. The binder is optionally at least one of polyvinylidene fluoride (PVDF), polytetrafluoroethylene (PTFE), polyacrylic acid (PAA), polyvinyl alcohol (PVA), ethylene-vinyl acetate copolymer (EVA), styrene butadiene rubber (SBR), carboxymethyl cellulose (CMC), sodium alginate (SA), polymethacrylic acid (PMA), and carboxymethyl chitosan (CMCS).

[0132] The positive electrode current collector may be made of a conductive carbon plate, a metal foil, a carbon-coated metal foil, a porous metal plate, or a composite current collector. The conductive carbon material of the conductive carbon plate is optionally one or more of super P, carbon black, Ketjen black, a carbon dot, a carbon nanotube, graphite, graphene, and a carbon nanofiber. The metal materials of the metal foil, the carbon-coated metal foil, and the porous metal plate are each independently selected from at least one of copper, aluminum, nickel, and stainless steel. The composite current collector may be a composite current collector formed by combining a

metal foil and a polymer base film.

[0133] In some embodiments, the positive electrode plate can be prepared in the following manner: dispersing the components described above for preparing the positive electrode plate, such as the positive electrode active material, the conductive agent, the binder, and any other components, in a solvent (such as N-methylpyrrolidone) to form a positive electrode slurry; and coating the positive electrode current collector with the positive electrode slurry, and performing drying, cold pressing, and other processes, such that the positive electrode plate can be obtained.

Separation Film

[0134] In some embodiments, the secondary battery further includes a separation film. The present application does not particularly limit the type of the separation film, and any porous-structure separation film known to have good chemical stability and mechanical stability may be selected and used.

[0135] In some embodiments, the material of the separation film may be selected from at least one of polyethylene, polypropylene, polyvinylidene fluoride, aramid, polyethylene terephthalate, polytetrafluoroethylene, polyacrylonitrile, polyimide, polyamide, polyester, and a natural fiber. The separation film may be a single-layer film or a multi-layer composite film, and there is no particular limitation on this. When the separation film is a multi-layer composite film, the materials of the layers may be the same or different, and there is no particular limitation on this.

[0136] In some embodiments, the positive electrode plate, the negative electrode plate, and the separation film may be manufactured into an electrode assembly through a winding process or a stacking process.

[0137] In some embodiments, the secondary battery may include an outer packaging. The outer packaging can be used for packaging the electrode assembly and electrolyte described above.

[0138] In some embodiments, the outer packaging of the secondary battery may be a hard shell, such as a hard plastic shell, an aluminum shell, or a steel shell. The outer packaging of the secondary battery may also be a soft pack, such as a pouch-type soft pack. The soft pack may be made of plastic, and examples of the plastic may include polypropylene, polybutylene terephthalate, polybutylene succinate, and the like.

Secondary Battery

[0139] A secondary battery includes a positive electrode plate, where the positive electrode plate includes the positive electrode active material according to some embodiments or the positive electrode active material prepared by the preparation method according to some embodiments.

[0140] In some embodiments, the secondary battery further includes a negative electrode plate, a separation film, and an electrolytic solution.

[0141] In some embodiments, the secondary battery is a negative electrode-free sodium battery.

[0142] In the negative electrode-free sodium battery, no negative electrode active material is pre-deposited, and only a negative electrode current collector is included. During the first charge, sodium ions gain electrons at the positive electrode side and are deposited as sodium metal on the surface of the current collector to form a sodium metal phase. During discharge, the sodium metal can be converted into sodium ions and return to the positive electrode, thus achieving cyclic charge and discharge. Compared to other sodium secondary batteries, the negative electrode-free sodium battery can achieve higher energy density due to the absence of limitations imposed by the negative electrode material. There is insufficient sodium metal in the negative electrode-free sodium battery as a negative electrode material to provide sufficient sodium element to the battery. Therefore, the application of the positive electrode active material provided in the present application in the negative electrode-free sodium battery can more effectively improve the cycle performance and high-rate capability of the battery.

[0143] In some embodiments, the CB value of the negative electrode-free sodium battery is less than or equal to 0.1.

[0144] The CB value is defined as the ratio of the unit area capacity of the negative electrode plate

to the unit area capacity of the positive electrode plate in a secondary battery. Since the negative electrode-free battery includes no negative electrode active material, the unit area capacity of the negative electrode plate is relatively small, and the CB value of the secondary battery is less than or equal to 0.1.

[0145] In some embodiments, a metal foil or a composite current collector may be used as the negative electrode current collector. For example, as the metal foil, an aluminum foil or a copper foil may be used. The composite current collector may include a polymer material substrate and a metal layer formed on at least one surface of the polymer material substrate. The composite current collector can be fabricated by forming a metal material (copper, copper alloy, nickel, nickel alloy, titanium, titanium alloy, silver, silver alloy, etc.) on a polymer material substrate (such as polypropylene (PP), polyethylene terephthalate (PET), polybutylene terephthalate (PBT), polystyrene (PS), and polyethylene (PE)).

[0146] In some embodiments, the negative electrode plate includes a negative electrode current collector and a bottom coating arranged on at least one surface of the negative electrode current collector, where the bottom coating includes one or more of a carbon nanotube, graphite, graphene, a silver-carbon composite nanoparticle, and a tin-carbon composite nanoparticle.

[0147] The bottom coating described above not only has excellent electrical conductivity, but is also conducive to the uniform deposition of metal ions on the surface of the negative electrode current collector of the negative electrode-free sodium battery, thereby improving the cycle performance and safety of the battery.

[0148] In some embodiments, the surface density of the bottom coating is 5 g/m.² to 50 g/m.².

[0149] In some embodiments, the surface density of the bottom coating is optionally 5 g/m.², 10 g/m.², 15 g/m.², 20 g/m.², 25 g/m.², 30 g/m.², 35 g/m.², 40 g/m.², 45 g/m.², or 50 g/m.².

[0150] In some embodiments, the thickness of the bottom coating is 2 μm to 100 μm.

[0151] In some embodiments, the thickness of the bottom coating is optionally 2 μm, 5 μm, 10 μm, 15 μm, 20 μm, 25 μm, 30 μm, 35 μm, 40 μm, 45 μm, 50 μm, 55 μm, 60 μm, 65 μm, 70 μm, 75 μm, 80 μm, 85 μm, 90 μm, 95 μm, or 100 μm.

[0152] The bottom coating described above has a low surface density and is thin, and cannot be used as a negative electrode active material. However, it is used as a bottom coating in a negative electrode-free sodium battery to reduce the nucleation overpotential of sodium, which is conducive to the uniform deposition of metal ions and inhibiting dendrite formation.

[0153] The present application does not particularly limit the shape of the secondary battery, and it may have a cylindrical shape, a prismatic shape, or any other shape. For example, FIG. 1 shows a secondary battery 5 having a prismatic structure as one example, and FIG. 2 shows an exploded view of the secondary battery 5.

[0154] In some embodiments, referring to FIG. 2, the outer packaging may include a housing 51 and a cover plate 53. The housing 51 may include a bottom plate and a side plate connected to the bottom plate, and the bottom plate and the side plate define, in an enclosing manner, an accommodating cavity. The housing 51 is provided with an opening communicating with the accommodating cavity, and the cover plate 53 is capable of lidding the opening to close the accommodating cavity. The positive electrode plate, the negative electrode plate, and the separation film may be subjected to a winding process or a stacking process to form an electrode assembly 52. The electrode assembly 52 is packaged in the accommodating cavity. The non-Newtonian fluid electrolyte composition is infiltrated into the electrode assembly 52. The number of the electrode assembly 52 included in the secondary battery 5 may be one or more, and those skilled in the art can select the number according to specific and actual needs.

Battery Module

[0155] In some embodiments, the secondary battery may be assembled into a battery module. The

number of secondary batteries included in the battery module may be one or more, and the specific number may be selected by those skilled in the art based on the use and capacity of the battery module.

[0156] FIG. 3 shows a battery module 3 as one example. Referring to FIG. 3, in the battery module 4, a plurality of secondary batteries 5 may be sequentially arranged in a length direction of the battery module 4. Certainly, the arrangement may also be in any other manner. Further, the plurality of secondary batteries 5 may be fixed by a fastener.

[0157] Optionally, the battery module 4 may further include a shell having an accommodating space in which the plurality of secondary batteries 5 are accommodated.

Battery Pack

[0158] In some embodiments, the battery module described above may also be assembled into a battery pack. The number of the battery modules included in the battery pack may be one or more, and the specific number may be selected by those skilled in the art based on the use and capacity of the battery pack.

[0159] FIG. 4 and FIG. 5 show a battery pack 1 as one example. Referring to FIG. 4 and FIG. 5, the battery pack 1 may include a battery case and a plurality of battery modules 4 arranged in the battery case. The battery case includes an upper case body 2 and a lower case body 3. The upper case body 2 is capable of lidding the lower case body 3 to form a closed space for accommodating the battery modules 4. The plurality of battery modules 4 may be arranged in any manner in the battery case.

Electric Device

[0160] In one embodiment of the present application, provided is an electric device, which includes at least one of the secondary battery according to any embodiment, the battery module according to any embodiment, or the battery pack according to any embodiment.

[0161] The electric device includes at least one of the secondary battery, the battery module, or the battery pack provided in the present application. The secondary battery, the battery module, or the battery pack can be used as a power source for the electric device, and they can also be used as an energy storage unit for the electric device. The electric device may include, but is not limited to, a mobile device (e.g., a mobile phone or a laptop computer), an electric vehicle (e.g., a pure electric vehicle, a hybrid electric vehicle, a plug-in hybrid electric vehicle, an electric bicycle, an electric scooter, an electric golf cart, or an electric truck), an electric train, ship, or satellite, an energy storage system, or the like.

[0162] As the electric device, a secondary battery, a battery module, or a battery pack may be selected based on its use requirements.

[0163] FIG. 6 shows an electric device as one example. The electric device is a pure electric vehicle, a hybrid electric vehicle, a plug-in hybrid electric vehicle, or the like. To meet the requirements of the electric device for high power and high energy density of the secondary battery, the battery pack or the battery module may be used.

[0164] As another example, the device may be a mobile phone, a tablet computer, a laptop computer, or the like. The device is generally required to be light and thin, and a secondary battery can thus be used as a power source.

EXAMPLES

[0165] Hereinafter, examples of the present application are described. The examples described below are illustrative and are merely used to explain the present application, and they should not be construed as limiting the present application. Examples without techniques or conditions specified therein are implemented according to techniques or conditions described in the literature in the art or according to product instructions. Reagents or instruments used herein without specified manufacturers are all commercially available conventional products.

I. Preparation Method

Example 1

1) Preparation of Positive Electrode Active Materials

[0166] 0.1 mol sodium pyrophosphate, 0.3 mol ferrous oxalate, 0.2 mol ammonium dihydrogen phosphate, 0.01 mol glucose, and 0.005 mol aluminum nitrate were added to 200 mL of deionized water, and the mixture was ground by a sand mill for 10 h to obtain a mixed slurry. The mixed slurry was dried by a spray drying device to obtain a dry precursor powder. The precursor was placed in a tube furnace, and nitrogen was introduced as a protective gas. The first calcination step was carried out by heating to 300° C. and holding for 4 h, followed by the second calcination step by heating to 550° C. and holding for 12 h, and a positive electrode active material was obtained.

2) Preparation of Positive Electrode Plate

[0167] 10 wt % polyvinylidene fluoride binder was thoroughly dissolved in N-methylpyrrolidone (NMP), and 10 wt % carbon black conductive agent and 80 wt % of the positive electrode active material described above were then added. The mixture was stirred and uniformly mixed to obtain a positive electrode slurry. The surface of the current collector aluminum foil was coated with the slurry evenly, and the current collector aluminum foil was then transferred to a vacuum drying oven for complete drying. The dried electrode plate was calendered and then punched to obtain a positive electrode plate.

3) Preparation of Negative Electrode Plate

[0168] The carbon nanotube and carboxymethyl cellulose (CMC) were added to deionized water in a mass ratio of 1:0.4, and the mixture was stirred to obtain a uniform slurry. The surface of the negative electrode current collector copper foil was coated with the slurry, and the negative electrode current collector copper foil was then transferred to a vacuum drying oven for complete drying and then punched. The bottom coating had a thickness of 20 μm and a surface density of 25 g/m². A negative electrode plate without a negative electrode structure was obtained.

4) Electrolytic Solution

[0169] In a glove box under argon atmosphere (H₂O<0.1 ppm, O₂<0.1 ppm), a sodium salt sodium hexafluorophosphate (NaPF₆) was dissolved in an organic solvent dimethoxyethane (DME), and the mixture was uniformly stirred to obtain an electrolytic solution with a sodium salt concentration of 1 mol/L.

5) Separation Film

[0170] A polypropylene film is used as the separation film.

6) Preparation of Button Battery

[0171] The button battery was assembled by using a sodium metal plate as a counter electrode, using a Celgard 2400 separation film, and injecting an electrolytic solution.

7) Preparation of Full Battery

[0172] The positive electrode plate, the separation film, and the negative electrode plate were stacked in sequence, with the separation film positioned between the positive electrode plate and the negative electrode plate to provide isolation. The electrolytic solution described above was then added, and a stacked battery was assembled.

[0173] The negative electrode-free sodium metal battery product of Example 1 was obtained.

Examples 2 to 6

[0174] The batteries of Examples 2 to 6 were prepared using a method similar to that of Example 1, except that the mass content of the carbon element in the positive electrode active material was adjusted. The specific parameters are shown in Table 1.

Example 7

[0175] The battery of Example 7 was prepared using a method similar to that of Example 1, except that the preparation method for the positive electrode active material was adjusted. The specific preparation method is as follows: [0176] 0.1 mol sodium pyrophosphate, 0.28 mol ferrous oxalate, 0.2 mol ammonium dihydrogen phosphate, 0.04 mol glucose, 0.02 mol manganese oxalate (M source), and 0.005 mol aluminum nitrate were added to 200 mL of deionized water, and the mixture was ground by a sand mill for 10 h to obtain a mixed slurry. The mixed slurry was dried by

a spray drying device to obtain a dry precursor powder. The precursor was placed in a tube furnace, and nitrogen was introduced as a protective gas. The system was heated to 300° C. for 4 h and then heated to 550° C. for 12 h, and a positive electrode active material was obtained.

Examples 8 to 12

[0177] The batteries of Examples 8 to 12 were prepared using a method similar to that of Example 1, except that the x value of the Mn element in the iron-based polyanionic compounds was adjusted. The specific parameters are shown in Table 1.

Examples 13 to 17

[0178] The batteries of Examples 13 to 17 were prepared using a method similar to that of Example 7, except that the mass content of the aluminum element in the positive electrode active material was adjusted. The specific parameters are shown in Table 1.

Examples 18 to 25

[0179] The batteries of Examples 18 to 25 were prepared using a method similar to that of Example 7, except that the temperature and the time of the second calcination step were adjusted, respectively. The specific parameters are shown in Table 1.

Examples 26 to 31

[0180] The batteries of Examples 26 to 31 were prepared using a method similar to that of Example 7, except that the type and the combination of transition metal elements in the iron-based polyanionic compounds were adjusted. The specific parameters are shown in Table 1.

Comparative Example 1

[0181] The battery of Comparative Example 1 was prepared using a method similar to that of Example 1, except that the preparation method of the positive electrode active material was adjusted. The preparation method is as follows: [0182] 0.1 mol sodium pyrophosphate, 0.3 mol ferrous oxalate, and 0.2 mol ammonium dihydrogen phosphate were added to 200 ml of water, and the mixture was ground by a sand mill for 10 h to obtain a mixed slurry. The mixed slurry was dried by a spray drying device to obtain a dry precursor powder. The precursor was placed in a tube furnace, and nitrogen was introduced as a protective gas. The first calcination step was carried out by heating to 300° C. and holding for 4 h, followed by the second calcination step by heating to 550° C. and holding for 12 h, and a positive electrode active material was obtained.

Comparative Example 2

[0183] The battery of Comparative Example 2 was prepared using a method similar to that of Comparative Example 1, except that a carbon material was introduced into the positive electrode active material. The specific parameters are shown in Table 1.

Comparative Example 3

[0184] The battery of Comparative Example 3 was prepared using a method similar to that of Comparative Example 2, except that an aluminum element was introduced into the positive electrode active material. The specific parameters are shown in Table 1.

Comparative Examples 4 to 5

[0185] The batteries of Comparative Examples 4 to 5 were prepared using a method similar to that of Comparative Example 1, except that Mn elements with different x values were introduced into the positive electrode active material. The specific parameters are shown in Table 1.

II. Performance Test

1. Performance Test of Positive Electrode Active Material

1) Test of Aluminum Element Content

[0186] The composition of the positive electrode active material was determined through inductively coupled plasma (ICP) spectroscopy analysis. For example, reference can be made to the standard YS/T 1006.2-2014, GB/T 23367.2-2009, or YS/T 1028.5-2015. Specifically, an inductively coupled plasma optical emission spectrometer (Thermo, ICAP7400; PerkinElmer, Avio200) can be used to measure the emission intensity of the aluminum element in the test sample. Standard samples with different contents of aluminum element were prepared, and their

emission intensities were measured to establish a standard curve. The mass content of the aluminum element in the test sample was quantified by matching its emission intensity to the standard curve.

2) Test of Residual Alkali Content

[0187] The residual alkali content of the prepared positive electrode active material was tested. In the present application, the residual alkali content of NaHCO_3 refers to the mass content of the measured mass of NaHCO_3 in the positive electrode active material. The residual alkali content was determined through a potentiometric titration method using chemical reagents. For example, reference can be made to the standard GB/T 9725-2007. Specifically, a Metrohm 905 Titrand potentiometric titrator can be used. The positive electrode active material was dissolved in deionized water, and the aqueous positive electrode active material solution was titrated with a standard titration solution. For each addition of 0.1 mL of the standard titration solution, the potential or pH value was recorded. The titration was stopped when the potential or pH value showed slight variation. The volume of the added standard titration solution and the measured potential or pH value were recorded, the titration end point was determined using a graphical method or the second derivative method, and the titration volume of the standard titration solution was determined. The mass of NaHCO_3 obtained through calculation was divided by the mass of the positive electrode active material, and the mass content was taken as the residual alkali content of NaHCO_3 in the positive electrode active material.

2. Test of Full Battery Performance

1) Specific Capacity Test of Button Battery

[0188] The process for testing the specific capacity of the button battery was as follows: At 25° C. and normal pressure, the button battery was charged at a constant current rate of 0.1 C to a voltage of 4 V, and then charged at a constant voltage of 4 V until the current dropped to 0.05 C, and the charge-specific capacity at this time was recorded as the first desodiation capacity. Subsequently, the battery was discharged at a constant current rate of 0.1 C to a voltage of 1.5 V, and the discharge-specific capacity at this time was recorded as the first sodiation capacity. The specific capacity of the positive electrode active material was the first sodiation capacity.

2) Capacity Retention Rate Test of the Full Battery After Cycling

[0189] The process for testing the capacity retention rate of the full battery was as follows: At 25° C., the prepared battery was charged at a constant current of 1 C to 3.7 V, then charged at a constant voltage of 3.7 V until the current dropped to 0.05 C, and then discharged at 1 C to 1.5 V. The obtained capacity was recorded as the initial capacity (C_0). The steps described above were repeated for the same battery described above, and the discharge capacity (C_n) of the battery after the nth cycle was recorded at the same time. The battery capacity retention rate P_n after each cycle was calculated as $P_n = C_n / C_0 \times 100\%$. A curve of the battery capacity retention rate versus the number of cycles was obtained, with the 200 point values of $P_1, P_2, \dots$, and 200 as the vertical coordinate and the corresponding cycle numbers as the horizontal coordinate. In this test process, the first cycle corresponded to $n=1$, the second cycle corresponded to $n=2, \dots$, and the 200th cycle corresponded to $n=200$. The capacity retention rate data of the battery corresponding to the examples and comparative examples in Table 1 were measured after 200 cycles under the test conditions described above, that is, the value of P_{200} . The test process of the comparative examples and other examples was the same as above.

3) Test of Rate Capability

[0190] The process for testing the rate capability was as follows: At 25° C., the prepared battery was placed in a constant temperature chamber at 25° C. and left to stand for 30 min to reach a constant temperature. After reaching the constant temperature, at 25° C., the battery was charged at a constant current of 0.33 C to 3.7 V, and then charged at a constant voltage of 3.7 V to 0.05 C; the battery was left to stand for 5 min, and then discharged at a constant current of 0.33 C to 1.5 V; the battery was left to stand for 5 min to obtain the discharge capacity C_1 at 0.33 C; subsequently, the

battery was charged at a constant current of 0.33 C to 3.7 V, and then charged at a constant voltage of 3.7 V to 0.05 C; the battery was left to stand for 5 min, and then discharged at a constant current of 3 C to 1.5 V; the battery was left to stand for 5 min to obtain the discharge capacity C2 at 3 C. The capacity retention rate at 3C rate was then calculated as $R=C2/C1 \times 100\%$. The test process of the comparative examples and other examples was the same as above.

III. Analysis of Test Results of Examples and Comparative Examples

[0191] The batteries of the examples and comparative examples were prepared according to the methods described above, and various performance parameters were measured. The results are shown in Table 1 below.

TABLE-US-00001	TABLE 1	Positive electrode active material	Battery Al	Carbon Capacity	element material	Temperature	Time of Residual	retention Capacity	mass mass of second	second
alkali	Specific rate	retention	Transition x	content/	content/	calcination	calcination	content/	capacity/	
after 200	rate at 3 C	No.	metal M	Value %	% step/°	C.	step/h	% (mAh/g)	cycles	rate
Example 1	/ 0	0.25	0.50	550	12	0.89	113.6	96.70%	90.8%	Example 2
Example 3	/ 0	0.25	2.10	550	12	0.90	116.4	97.8%	91.4%	Example 4
Example 5	/ 0	0.25	4.50	550	12	0.91	111.5	98.60%	92.9%	Example 6
Example 7	Mn	0.2	0.25	2.1	550	12	0.89	115.7	98.0%	Example 8
Example 9	Mn	0.1	0.25	2.1	550	12	0.88	114.2	97.5%	Example 10
Example 11	Mn	0.4	0.25	2.1	550	12	0.87	109.5	96.4%	Example 12
Example 13	Mn	0.2	0.03	2.1	550	12	1.13	111.3	96.7%	Example 14
Example 15	Mn	0.2	0.4	2.1	550	12	0.83	110.9	96.3%	Example 16
Example 17	Mn	0.2	0.9	2.1	550	12	0.61	102.1	93.2%	Example 18
Example 19	Mn	0.2	0.25	2.1	525	12	0.97	114.3	97.6%	Example 20
Example 21	Mn	0.2	0.25	2.1	600	12	0.68	109.4	97.4%	Example 22
Example 23	Mn	0.2	0.25	2.1	550	8	0.78	111.4	96.50%	Example 24
Example 25	Mn	0.2	0.25	2.1	550	14	0.96	113.5	97.10%	Example 26
Example 27	Mn + Ni	0.2	0.25	2.1	550	12	0.89	112.7	97.3%	Example 28
Example 29	Mn + Ni + Co	0.2	0.25	2.1	550	12	0.90	112.6	97.1%	Example 30
Example 31	Co	0.2	0.25	2.10	550	12	0.90	114.2	97.0%	Comparative
Example 1	Comparative	///	2.10	550	12	1.86	96.5	95.20%	87.70%	Example 2
Example 3	Comparative	Mn	0.1	//	550	12	1.90	91.2	84.40%	Example 4
Example 5	Comparative	Mn	0.3	//	550	12	1.93	88.6	83.80%	Example 5

[0192] According to the results described above, Examples 1 to 31 all involve positive electrode active materials for use in a secondary battery, which includes an iron-based polyanionic compound and an aluminum-containing oxide, where the iron-based polyanionic compound has the following general formula: $\text{Na}_{0.4}\text{Fe}_{0.3-x}\text{M}_x\text{Al}_y(\text{PO}_4)_2\text{P}_2\text{O}_7/\text{C}$, where M includes a transition metal element, $0 \leq x \leq 0.5$, and $0 < y < 0.2$.

[0193] It can be seen from the comparison of Examples 1 to 31 with Comparative Examples 1 to 2 that the doping of an Al element in the iron-based polyanionic compound and the residue of an Al source-derived oxide on the surface of the iron-based polyanionic compound effectively reduce the residual alkali content of the positive electrode active material increase the specific capacity of the positive electrode active material, and increase the capacity retention rate of the battery at a high rate.

[0194] It can be seen from the comparison of Examples 1 to 31 with Comparative Example 3 that controlling the y value in the positive electrode active material to be greater than 0 and less than 0.2

is conducive to increasing the specific capacity of the positive electrode active material, and increasing the capacity retention rate of the battery after 200 cycles and the capacity retention rate at 3C rate, thereby broadening the application of the battery.

[0195] It can be seen from the comparison of Examples 7 to 31 with Comparative Examples 4 to 5 that, compared with traditional positive electrode active materials containing only transition metal elements, the positive electrode active material including an aluminum-containing oxide and an Al-doped iron-based polyanionic compound is conducive to reducing the residual alkali content of the positive electrode active material, increasing the specific capacity of the positive electrode active material, and increasing the capacity retention rate of the battery after 200 cycles and the capacity retention rate at 3C rate.

[0196] It can be seen from the comparison of Examples 1 to 6 with Comparative Example 1 that controlling the mass content of the carbon material to be 0.5% to 6% based on the total mass of the positive electrode active material is conducive to increasing the specific capacity of the positive electrode active material and increasing the capacity retention rate of the battery after 200 cycles and the capacity retention rate at 3C rate. It can be seen from the comparison of Examples 2 to 4 with Examples 1 and 5 to 6 that controlling the mass content of the carbon material to be 1% to 3.5% based on the total mass of the positive electrode active material is conducive to further increasing the specific capacity of the positive electrode active material.

[0197] It can be seen from the comparison of Examples 7 to 12 with Comparative Examples 4 to 5 that the introduction of the carbon material, the doping of an Al element in the iron-based polyanionic compound, and the residue of an Al source-derived oxide on the surface of the iron-based polyanionic compound effectively reduce the residual alkali content of the positive electrode active material, increase the specific capacity of the positive electrode active material, and increase the capacity retention rate of the battery after 200 cycles and the capacity retention rate at 3C rate.

[0198] It can be seen from the comparison of Examples 7 to 12 with Example 3 that the introduction of transition metal elements and controlling the x value to be less than or equal to 0.5 are conducive to further increasing the capacity retention rate of the battery at 3C rate.

[0199] It can be seen from the comparison of Examples 7 and 13 to 17 with Comparative Example 2 that the introduction of an aluminum element and an aluminum-containing oxide into the positive electrode active material and controlling the total mass content of the aluminum element in the positive electrode active material to be 0.03% to 0.9% based on the total mass of the positive electrode active material are conducive to reducing the residual alkali content of the positive electrode active material, increasing the specific capacity of the positive electrode active material, and increasing the capacity retention rate of the battery at 3C rate.

[0200] It can be seen from the comparison of Examples 7 and 18 to 21 with Comparative Example 2 that controlling the calcination temperature to be 500° C. to 600° C. is conducive to reducing the residual alkali content of the positive electrode active material, increasing the specific capacity of the positive electrode active material, and increasing the capacity retention rate of the battery after 200 cycles and the capacity retention rate at 3C rate. It can be seen from the comparison of Examples 7 and 19 to 20 with Examples 18 and 21 that controlling the calcination temperature to be 525° C. to 575° C. is conducive to further increasing the specific capacity of the positive electrode active material and increasing the capacity retention rate of the battery after 200 cycles and the capacity retention rate at 3C rate. It can be seen from the comparison between Examples 7 and 20 with Examples 18 to 19 and 21 that controlling the calcination temperature to be 550° C. to 575° C. is conducive to increasing the capacity retention rate of the battery after 200 cycles and the capacity retention rate at 3C rate to a greater extent.

[0201] It can be seen from the comparison of Examples 7 and 22 to 25 with Comparative Example 2 that controlling the calcination time to be 8 h to 15 h is conducive to reducing the residual alkali content of the positive electrode active material, increasing the specific capacity of the positive electrode active material, and increasing the capacity retention rate of the battery after 200 cycles

and the capacity retention rate at 3C rate. It can be seen from the comparison of Examples 7 and 23 to 24 with Comparative Examples 22 and 25 that controlling the calcination time to be 10 h to 14 h is conducive to further increasing the specific capacity of the positive electrode active material and increasing the capacity retention rate of the battery after 200 cycles and the capacity retention rate at 3C rate.

[0202] It can be seen from the comparison of Examples 7 and 26 to 31 with Comparative Examples 4 to 5 that controlling the transition metal elements in the positive electrode active material to include one or more of Ni, Co, Mn, and Ca is conducive to reducing the residual alkali content of the positive electrode active material, increasing the specific capacity of the positive electrode active material, and increasing the capacity retention rate of the battery after 200 cycles and the capacity retention rate at 3C rate.

[0203] It can be seen from the comparison of Example 26 with Examples 7 and 29, as well as Example 27 with Examples 7 and 30 that controlling the transition metal elements in the positive electrode active material to include two of Ni, Co, and Mn, rather than only one of Ni, Co, and Mn, is conducive to further increasing the capacity retention rate of the battery at 3C rate. It can be seen from the comparison of Example 28 with Examples 26 to 27 that controlling the transition metal elements in the positive electrode active material to include three of Ni, Co, and Mn, rather than only one of Ni, Co, and Mn, is conducive to increasing the capacity retention rate of the battery at 3C rate to a greater extent.

[0204] It should be noted that the present application is not limited to the embodiments described above. The embodiments described above are merely examples, and any embodiments having a structure substantially identical to the technical concept and exerting the same functional effects within the scope of the technical solutions of the present application are all included within the technical scope of the present application. Furthermore, without departing from the spirit of the present application, various modifications that can be conceived by those skilled in the art to the embodiments, as well as other embodiments formed by combining some of the constituent elements of the embodiments, are also included within the scope of the present application.

Claims

1. A positive electrode active material for use in a secondary battery, wherein the positive electrode active material comprises a carbon material-composited iron-based polyanionic compound and an aluminum-containing oxide, and the iron-based polyanionic compound has the following formula: $\text{Na}_{0.4}\text{Fe}_{0.3-x}\text{M}_x\text{Al}_y(\text{PO}_4)_2\text{P}_2\text{O}_7/\text{C}$, wherein M comprises a transition metal element, $0 \leq x \leq 0.5$, and $0 < y < 0.2$.
2. The positive electrode active material according to claim 1, wherein the aluminum-containing oxide is distributed on at least a portion of the surface of a primary particle of the iron-based polyanionic compound.
3. The positive electrode active material according to claim 1, wherein a mass content of an aluminum element in the positive electrode active material is 0.03% to 0.9% based on a total mass of the positive electrode active material.
4. The positive electrode active material according to claim 1, wherein M comprises one or more of Ni, Co, Mn, Cu, V, Ti, Mo, Nb, W, Cr, Zn, Zr, and Ca.
5. The positive electrode active material according to claim 1, wherein M comprises at least two of Ni, Co, Mn, Cu, V, and Ca.
6. The positive electrode active material according to claim 1, wherein the surface of the iron-based polyanionic compound is coated with the carbon material in the form of a carbon film, or the carbon material is distributed between primary particles of the iron-based polyanionic compound in the form of particles.
7. The positive electrode active material according to claim 1, wherein the carbon material

comprises one or more of amorphous carbon, conductive carbon black, a carbon nanotube, and graphene.

8. The positive electrode active material according to claim 1, wherein a mass content of the carbon material is 0.5% to 6%, based on the total mass of the positive electrode active material.

9. The positive electrode active material according to claim 1, wherein a specific capacity of the positive electrode active material is not less than 98 mAh/g.

10. The positive electrode active material according to claim 1, wherein a residual alkali content of NaHCO_3 in the positive electrode active material is less than 1.2% based on the total mass of the positive electrode active material.

11. A preparation method for a positive electrode active material for use in a secondary battery, comprising the following: dissolving raw materials comprising an iron source, a sodium source, a phosphorus source, an aluminum source, and a carbon source in water to obtain a mixed slurry, wherein the raw materials optionally comprise an M source, and the M source is a salt containing a transition metal; and drying and then calcining the mixed slurry to prepare the positive electrode active material comprising an iron-based polyanionic compound and an aluminum-containing oxide, wherein the iron-based polyanionic compound has the following formula:

$\text{Na}_{0.4}\text{Fe}_{0.3-x}\text{M}_x\text{Al}_y(\text{PO}_4)_2\text{P}_2\text{O}_7/\text{C}$, wherein M comprises a transition metal element, $0 \leq x \leq 0.5$, and $0 < y < 0.2$.

12. The preparation method according to claim 11, wherein drying and then calcining the mixed slurry comprises the following: drying the mixed slurry to obtain a precursor powder; and subjecting the precursor powder to a stepwise calcination process, with a temperature of the first calcination step being 300° C. to 400° C. and a time being 3 h to 6 h, and a temperature of the second calcination step being 500° C. to 600° C. and a time being 8 h to 15 h, to prepare the positive electrode active material.

13. The preparation method according to claim 12, wherein the temperature of the second calcination step is 525° C. to 575° C.

14. The preparation method according to claim 12, wherein the time of the second calcination step is 10 h to 14 h.

15. The preparation method according to claim 11, wherein the M source comprises one or more of nickel nitrate, nickel acetate, cobalt nitrate, manganese nitrate, cobalt acetate, manganese acetate, nickel oxalate, cobalt oxalate, manganese oxalate, nickel oxide, cobalt oxide, and manganese oxide.

16. The preparation method according to claim 11, wherein the aluminum source comprises one or more of aluminum nitrate, aluminum acetylacetonate, aluminum acetate, aluminum hydroxide, and aluminum oxide.

17. The preparation method according to claim 11, wherein the carbon source comprises one or more of sucrose, tannic acid, polyethylene glycol, polyacrylonitrile, cellulose, polyvinylpyrrolidone, sucrose, oxalic acid, glucose, ascorbic acid, polyethylene, citric acid, conductive carbon black, a carbon nanotube, and graphene.

18. A secondary battery, comprising a positive electrode plate, wherein the positive electrode plate comprises the positive electrode active material according to claim 1.

19. The secondary battery according to claim 18, wherein the secondary battery comprises a sodium battery.

20. The secondary battery according to claim 18, wherein the secondary battery is a negative electrode-free sodium battery.
